

Pranav Mishra

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Technical Experience

Founding LLM Engineer | *alfred_ ↗* | New York City, NY **Apr 2026 - Present**

Early-stage AI startup, Founding team; multi-agent personal assistant over SMS & email, 5K+ subscribers

- Own the reliability layer of the multi-agent assistant: built the eval harness from scratch; production-scenario replay with hybrid deterministic + LLM-as-judge scoring, catching regressions across 12 failure classes in nightly replays
- Replaced per-message LLM calls in the email-rules engine with a deterministic three-stage matcher, cutting per-user LLM cost 40% at sub-50ms latency with parity match accuracy across 100K+ daily emails
- Re-architected agent memory to ground recall in verified state, cutting fabricated-recall & false-confirmation incidents 80%

AI Engineer | *WheelPrice ↗* | Charlotte, NC **Jul 2025 - Mar 2026**

Techstars-backed startup building a marketplace for automotive/aftermarket wheels; two-engineer team

- Owned a React / Node.js / MongoDB CMS serving 15K daily users; cut API p95 latency from 1.5s to 300ms via query-plan tuning, connection pooling, and Redis caching; added SSR and dynamic SEO, lifting organic traffic by 25%
- Shipped an XGBoost churn-prediction pipeline, hitting 0.85 AUC and flagging at-risk accounts weekly for the ops team

Research Software Engineer | *UIC, VARE Labs ↗* | Chicago, IL **Feb 2024 - Feb 2026**

Applied AI research lab, UIC Department of Applied Health Sciences; VR, voice, and clinical AI systems

- Deployed a quantized (INT8, TorchScript) audio ML inference service at 150ms p95 powering assistant-patient voice interaction; built the Python/PostgreSQL API backend, supporting 200 concurrent users across the UIC hospital network

Research & Publications

TeamMedAgents: SLM-based multi-agent medical reasoning *GitHub | Paper | Code-SLM*

- First-author(under review): multi-agent medical-reasoning system where small language models match frontier LLM accuracy (77.6% across 8 benchmarks) at $3.1\times$ inference speed via trust-weighted voting & tool-augmented retrieval

MetaRAG: Metadata Enrichment for RAG Systems *GitHub | IEEE Paper*

- First-author (IEEE CAI 2026, accepted): retrieval system (LLM metadata generation, cross-encoder reranking, hybrid embeddings) hitting 82.5% precision / 92.5% Hit@10 & 20% fewer hallucinations vs baseline RAG (AWS, p99 <300ms)

Education

Master of Science, Computer Science **Aug 2023 - May 2025**

University of Illinois at Chicago - Chicago, IL, USA | Graduate Assistant

Coursework: Advanced ML, Applied AI, Advanced NLP, Computer Vision, VR, Game Design, Object-Oriented Programming

Bachelor of Science, Computer Science & Engineering **Aug 2019 - Jun 2023**

Dayananda Sagar College of Engineering, Bangalore, Karnataka, India

Projects

MockFlow-AI: Real-Time Voice Interview Platform with Multi-Agent Architecture *GitHub | Website*

- Shipped a real-time voice interview platform: LiveKit, multi-agent orchestration, streaming STT/TTS, sub-400ms latency
- Architected an LLM-as-judge hiring-verdict engine: evidence-cited scoring across 5 competencies, deterministic verdict computation, calibrated against a 50-item human gold set (weighted Cohen's $\kappa = 0.82$)

SnakeAI: Multi-Agent Reinforcement Learning Snake Game *GitHub | Demo*

- Built a C++/LibTorch RL framework training DQN & PPO agents to convergence, with a 50K iteration soak & Docker CI/CD

SoulEngine: Autonomous Game-NPC Agent Framework with MCP Tool Orchestration *GitHub*

- Built a TypeScript framework for autonomous game NPCs: a planning/memory agent separated from the in-game persona, MCP tool invocation, & inter-NPC communication at sub-second latency, driving 20 concurrent NPCs in a live game scene

Skills & Extracurricular

TECHNICAL SKILLS: Python, TypeScript, C++, JavaScript, LangChain, MCP, Node.js, FastAPI, React, PostgreSQL, Redis, MongoDB, Docker, Git, CI/CD, AWS, Azure, PyTorch, LibTorch, Hugging Face

AGENTIC TOOLING: Claude Code + Cursor daily driver, custom subagents, hooks, parallel worktrees, agentic-loop design

Winner of MIT XR Hackathon | Built Meta Quest 3 app using Unity & Hugging Face for spatial data visualization ↗

INFORMS Analytics+ Speaker | Presented MetaRAG to 700+ professionals | First place, HINT 5.0 (interactive museum) ↗